

using the following commands. Copy the `amv-ffmpeg` that the following steps generate to the `/usr/bin` directory.

```
git clone https://github.com/vsubhash/amv-ffmpeg.git
cd amv-ffmpeg
./configure
make
```

So, what can FFmpeg do?

- **List supported codecs:** The `-codecs` or `-formats` options will list all the codecs that are supported. Each codec is referred to by a special name, which will have to be used in your FFmpeg commands when a particular codec needs to be used. The `-encoders` option will list all codecs which FFmpeg can convert to, i.e., the output formats. The `-decoders` option will list all codecs which FFmpeg can convert from, i.e., the input formats.
- **Check codecs used in a media file:** FFmpeg uses the `-i` option to specify the input file. This will display the streams available in the file and the settings with which each stream was encoded. Do remember that file extensions can be misleading. MP4, AVI or WMV files are merely containers for multiple content streams. There is no guarantee that a WMV file will have a stream in Windows Media Audio format. It could very well have MPEG-3 audio.

```
ffmpeg -i duffy.ogg
```

- **Convert a file from one format to another:** Linux programs typically create OGV/OGG files. However, very few consumer electronic devices support these formats. You can use FFmpeg to convert to a device-compatible format, as it can guess the output format based on the file extension you have used.

```
ffmpeg -i duffy.ogg duffy.mp4
```

- **Convert a file to AMV format:** First you have to convert the video to WMV format. Then, you can use the AMV FFMPEG binary.

```
ffmpeg -i duffy.ogg duffy.wmv
```

```
amv-ffmpeg -i duffy.wmv -f amv \
-s 160x120 -r 16 -ac 1 -ar 22050 -qmin 3 -qmax 3 \
tank.amv
```

- **Explicitly specify conversion settings:** When you let FFmpeg guess the output format and do the conversion, it will choose its own default codec settings. This may not always be what you want. Sometimes, you may have very different requirements. In this example, the container format is MP4. The video stream is set to the MPEG4 codec, 30 frames per second and a 500K bit rate.

The audio stream is set to LAME MP3 codec, a 44.1kHz sampling rate and a 96K bit rate. Here, `mpeg4` and `libmp3lame` are the codec names.

```
ffmpeg -i duffy.ogg \
-vcodect mpeg4 -s 352x480 -r 30 -b:v 500k \
-acodect libmp3lame -ar 44100 -b:a 96k \
duffy.mp4
```

- **Extract audio from a video file:** You can use the `-vn` option to disable the video stream and save the remaining audio stream to an MP3 file, as follows:

```
ffmpeg -i duffy.ogg \
-vn \
-acodect libmp3lame -ar 44100 -b:a 96k \
duffy.mp3
```

- **Help listen to YouTube and other online video sites:** I use the `youtube-dl` program to 'listen' to YouTube videos. Many TV programmes, such as the Keiser Report, upload their shows to YouTube so that viewers who miss the broadcast schedule can catch up later. With an audio file, you can continue to work while you listen. To install `youtube-dl`, use the command `sudo apt-get install youtube-dl`. The first `youtube-dl` command here uses the `-F` option (uppercase F) to list all formats in which a particular video is available. The second `youtube-dl` command uses the `-f` (lowercase F) to download the video in a particular format identified by a number obtained from the previous command. Later, FFmpeg is used to convert the audio to OGG format so that it can be played with the Audacious player. Audacious has an alternative Winamp interface, which can be minimised to a teeny-weeny but entirely functional floating toolbar. If you would rather view the video, then you can try the YouTube downloader JavaScript given in my article carried in the February 2016 issue of *OSFY*. You can choose that option if your Net connection does not support streaming videos very well.

```
# list the available formats
youtube-dl -F \
https://www.youtube.com/watch?v=_juHRSuodjU
```

```
# download the video in M4A audio format
youtube-dl -f 140 \
https://www.youtube.com/watch?v=_juHRSuodjU
```

```
ffmpeg -i *.m4a play.ogg
audacious play.ogg & # return to prompt
```

- **Fixe errors:** `recordMyDesktop` is a great screen-capturing program but it sometimes creates video files